

Lemon Bay Aquatic Preserve
SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

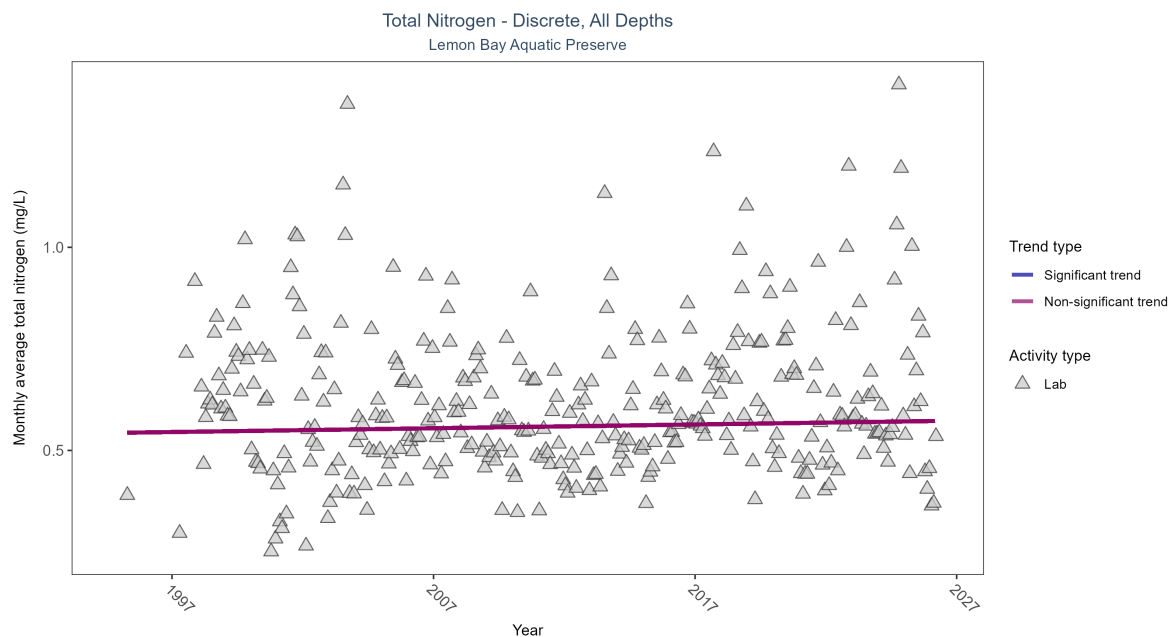


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 1995 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	3165	31	1995 - 2026	0.555	0.03785	0.54345	0.00093	0.3176

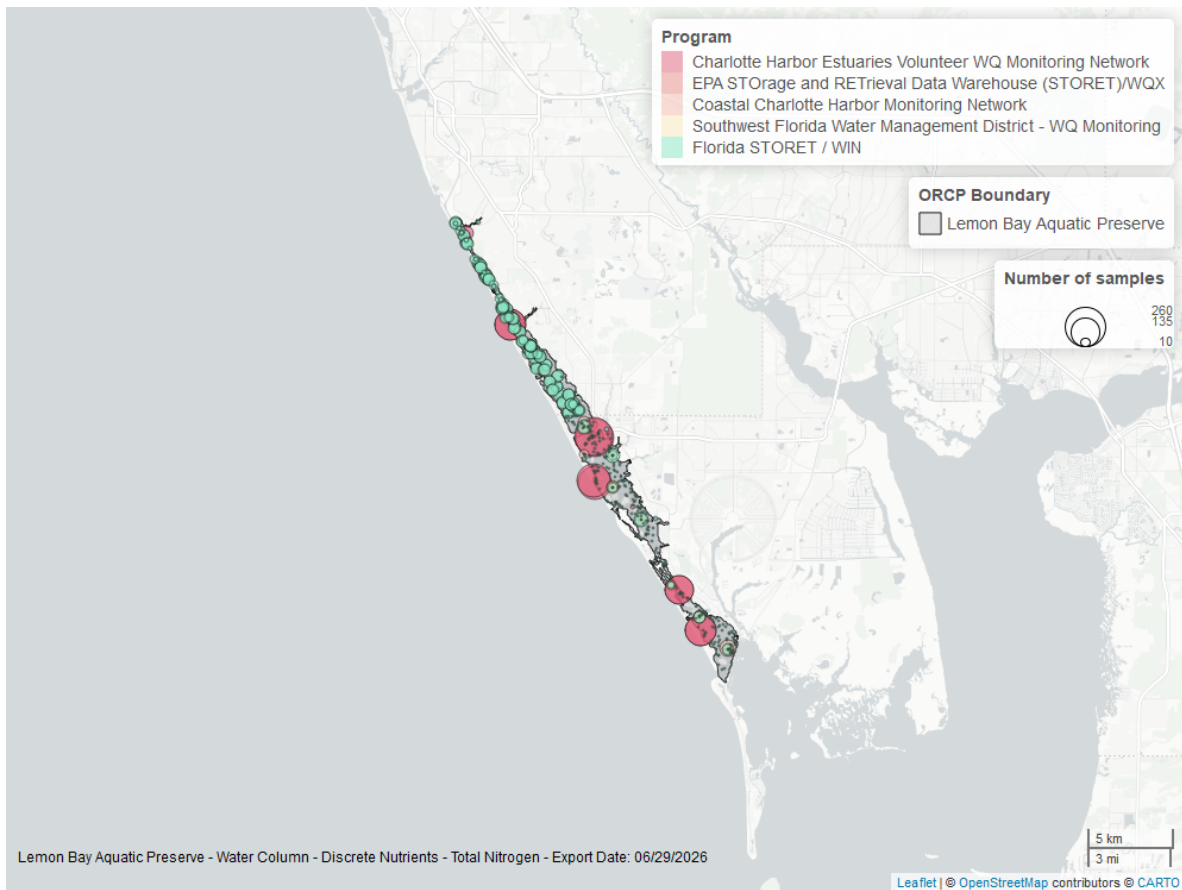


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

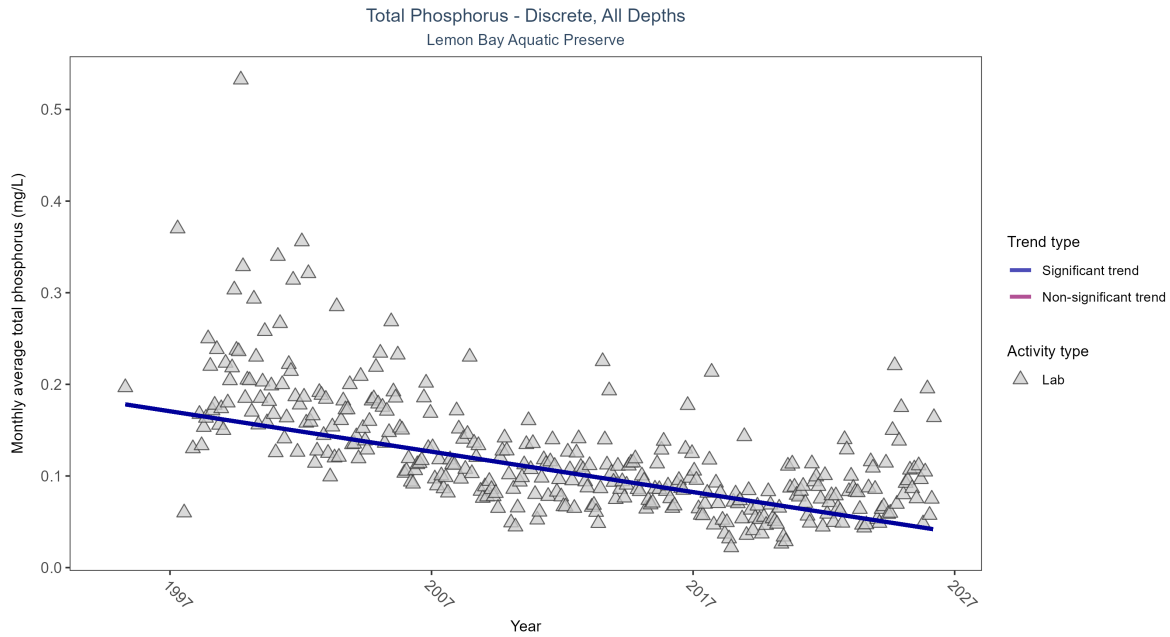


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	3211	31	1995 - 2026	0.083	-0.52037	0.17944	-0.00441	0

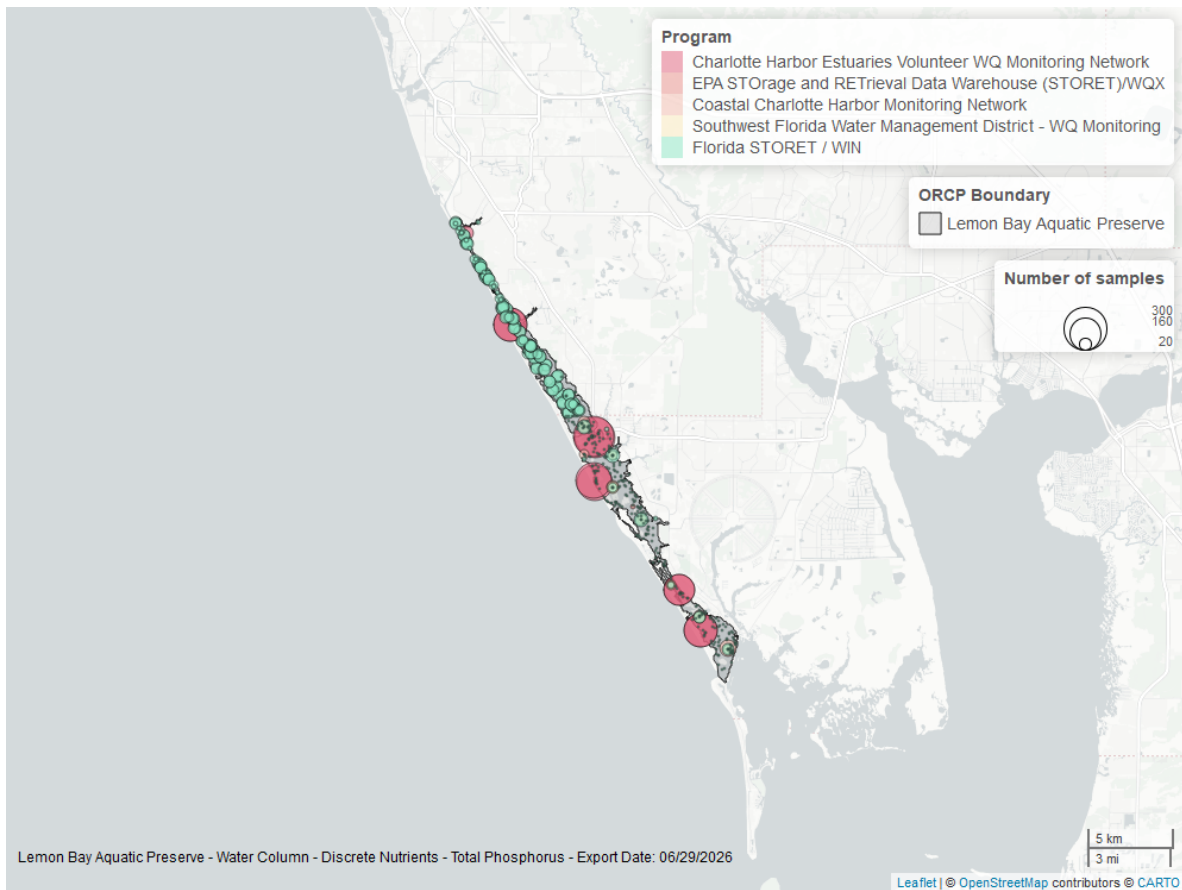


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Discrete

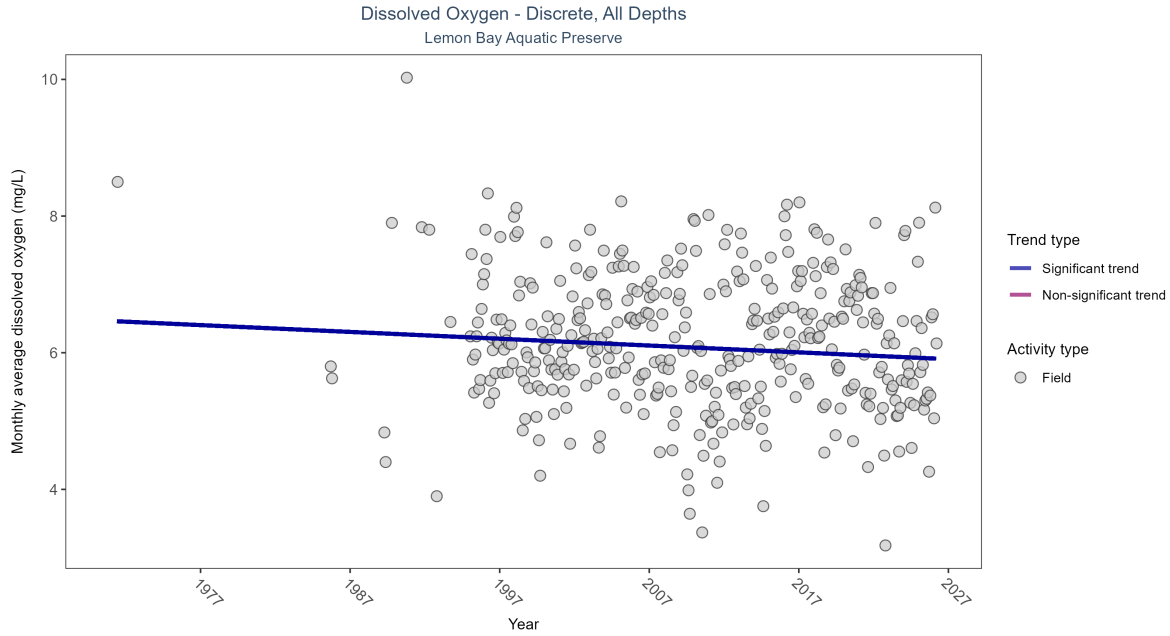


Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 3: Monthly average dissolved oxygen decreased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	13898	39	1971 - 2026	6.1	-0.07856	6.46405	-0.00997	0.0216

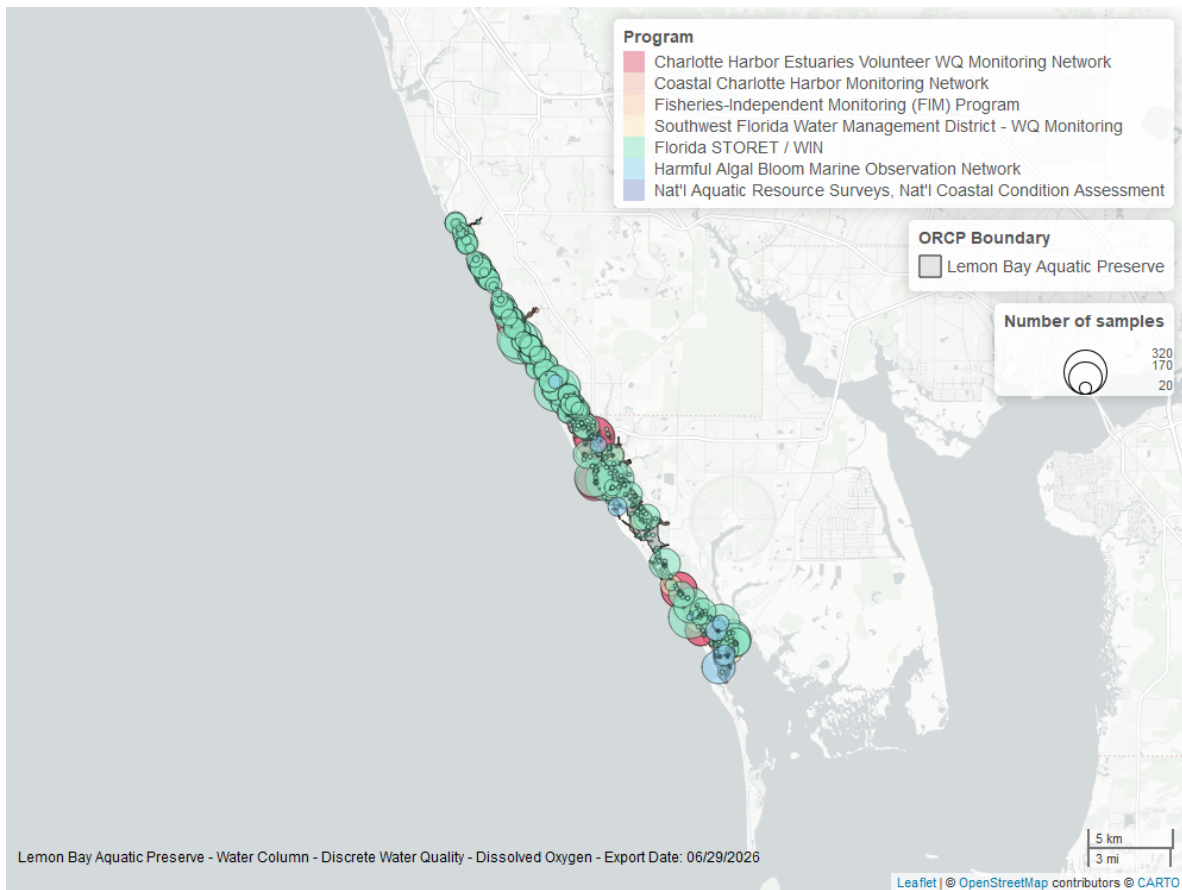


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

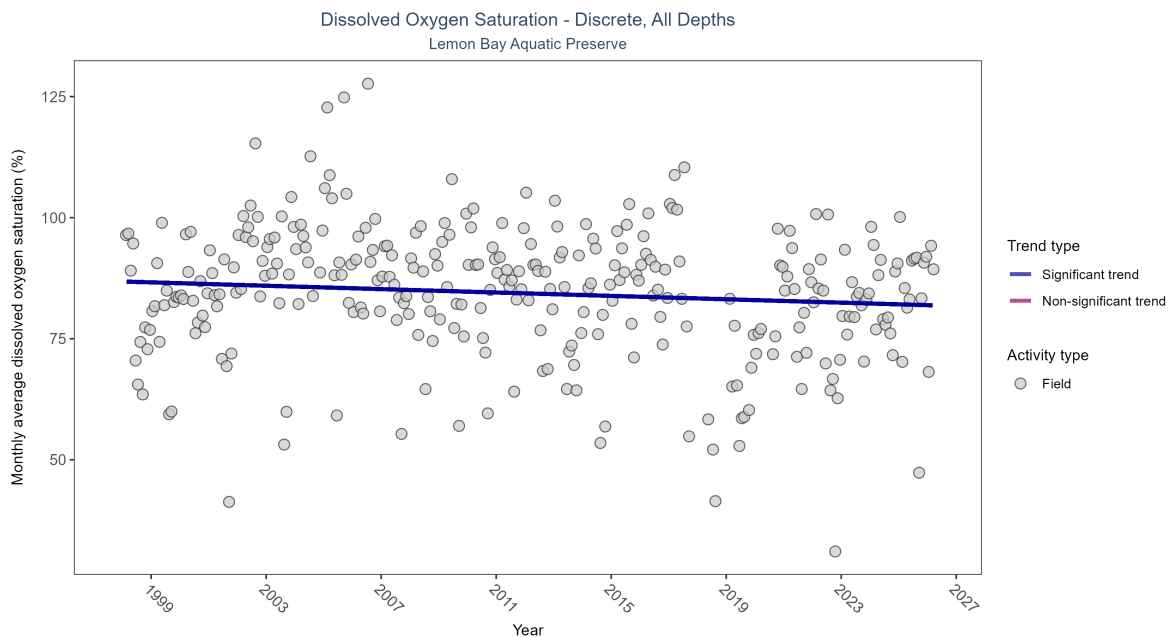


Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation decreased by 0.17% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	4738	29	1998 - 2026	87	-0.09436	86.81334	-0.17462	0.0182

\end{table}

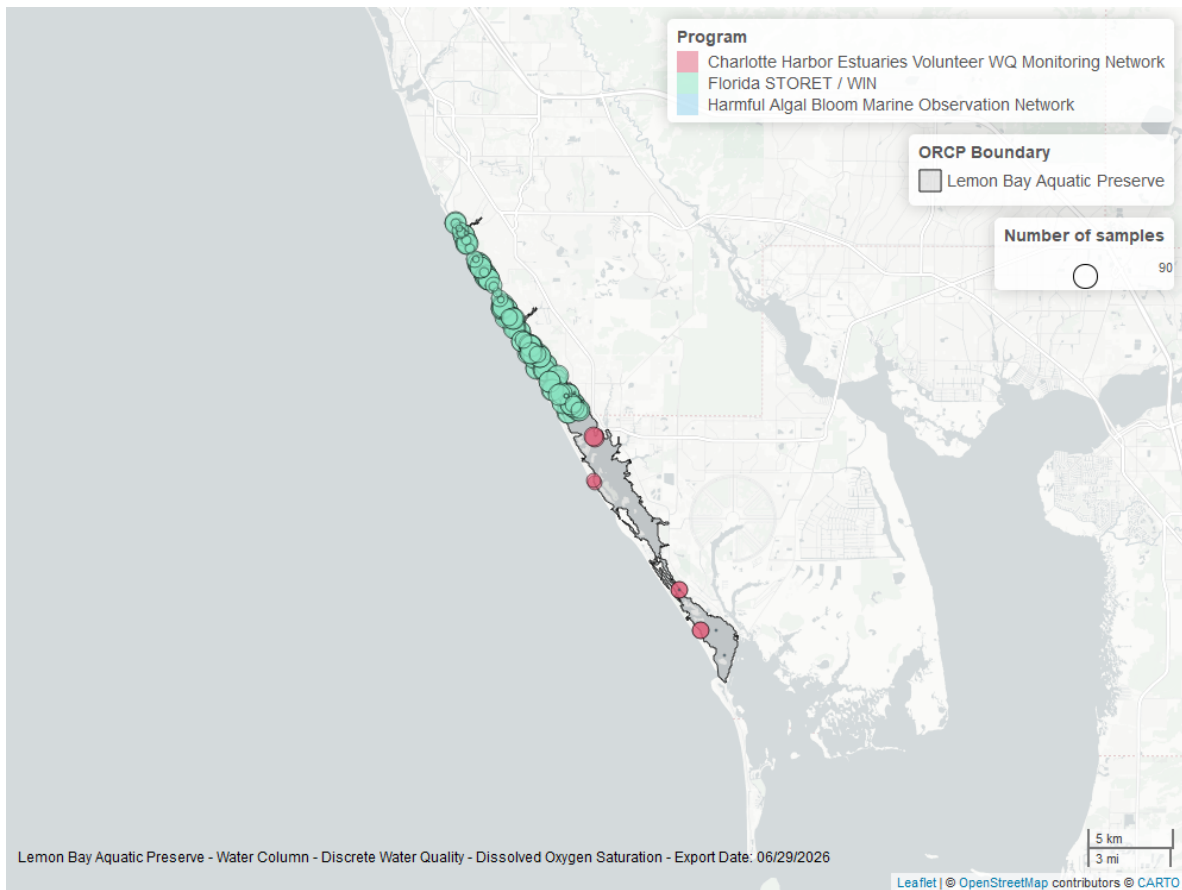


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

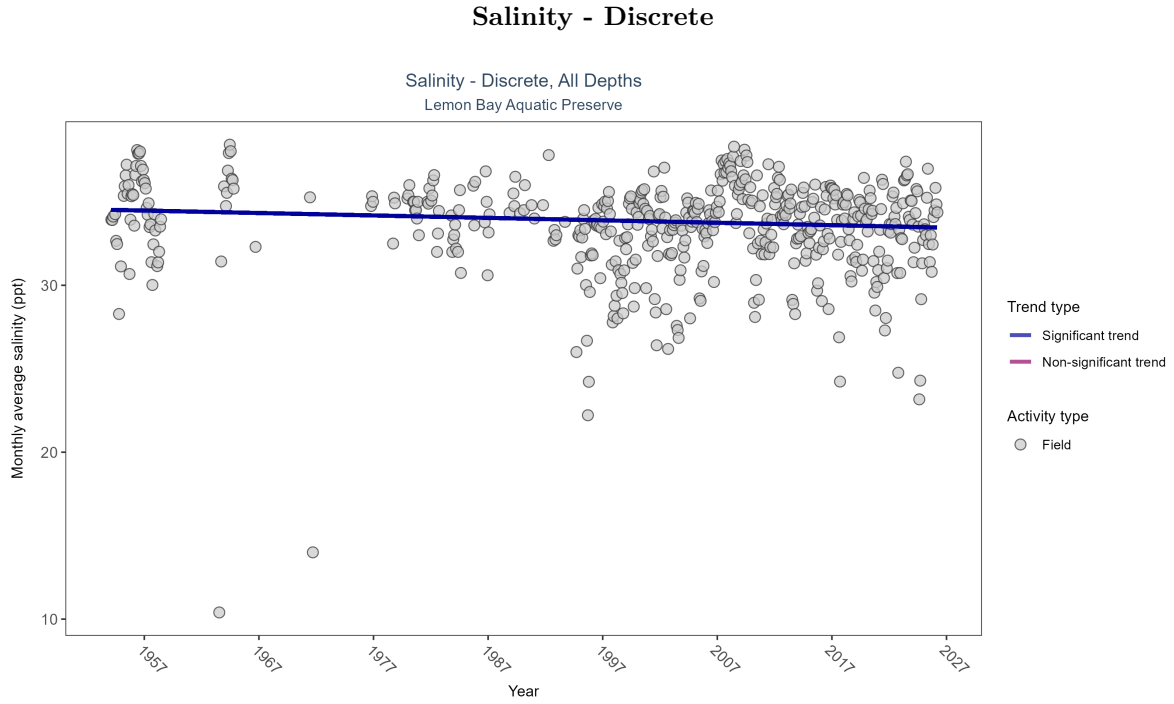


Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 4: Monthly average salinity decreased by 0.01 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	11609	59	1954 - 2026	34	-0.07872	34.51915	-0.01459	0.0065

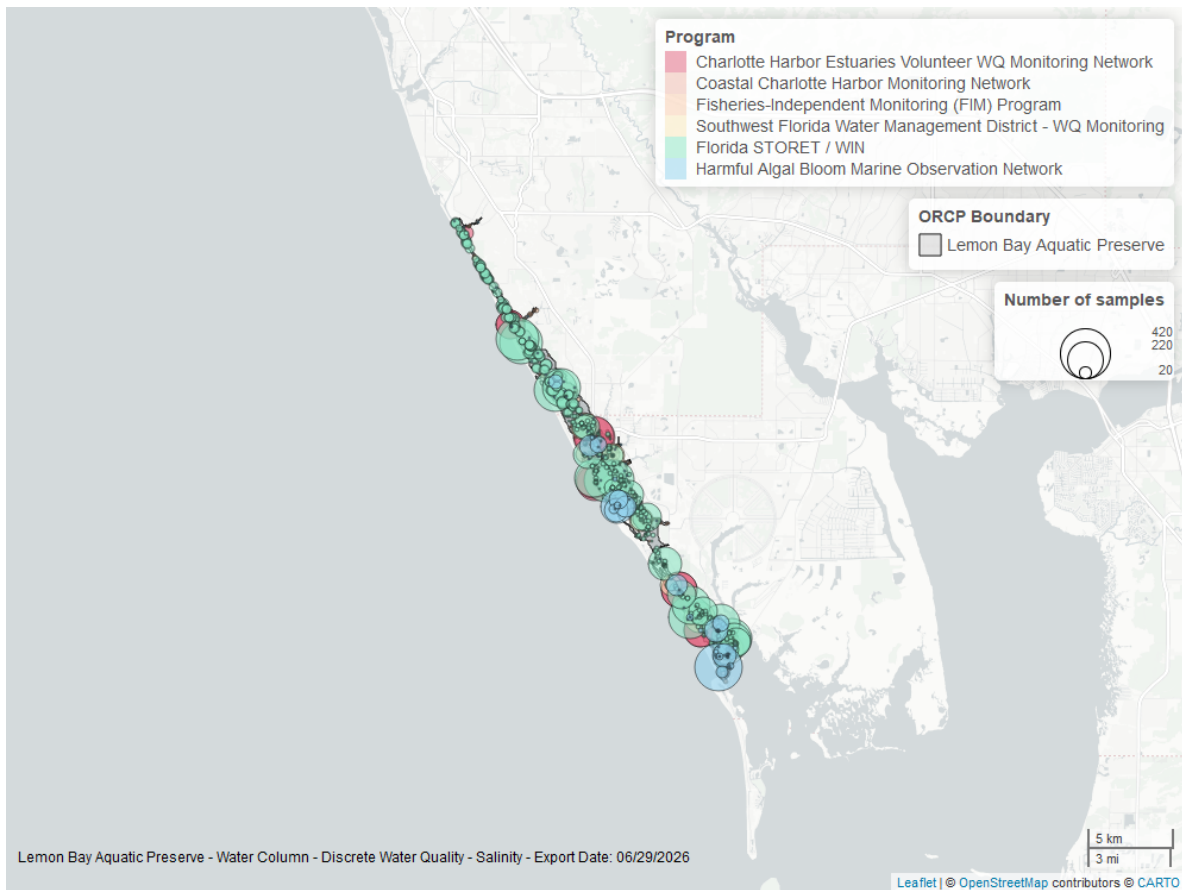


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

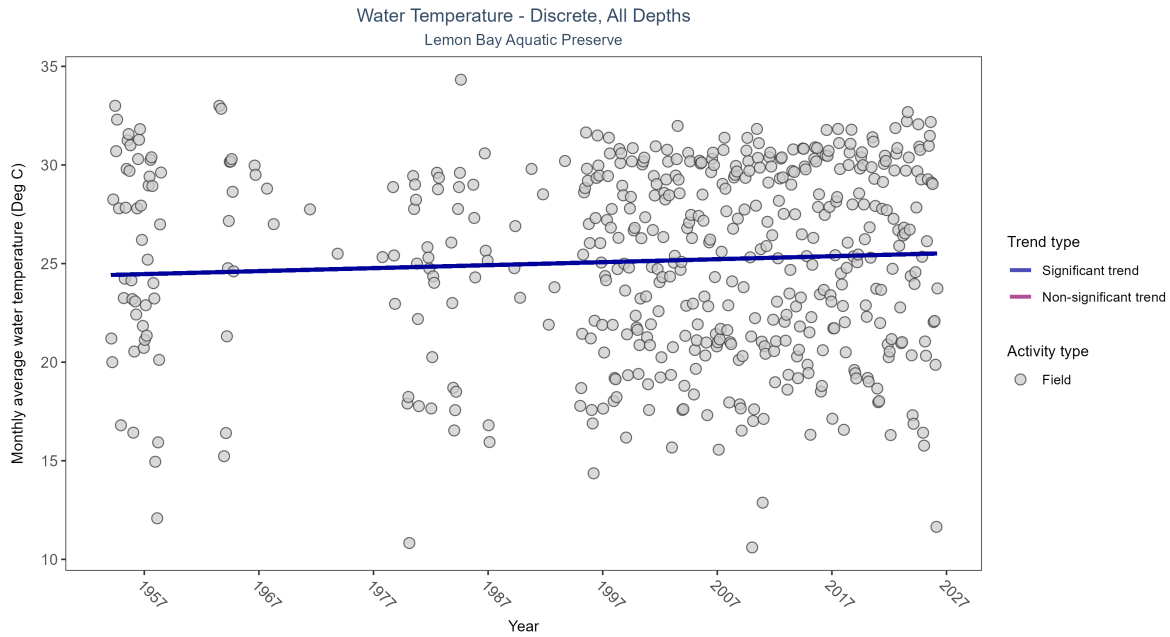


Figure 11: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 5: Monthly average water temperature increased by 0.02Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	15111	60	1954 - 2026	26.5	0.12148	24.42082	0.01512	0.0001

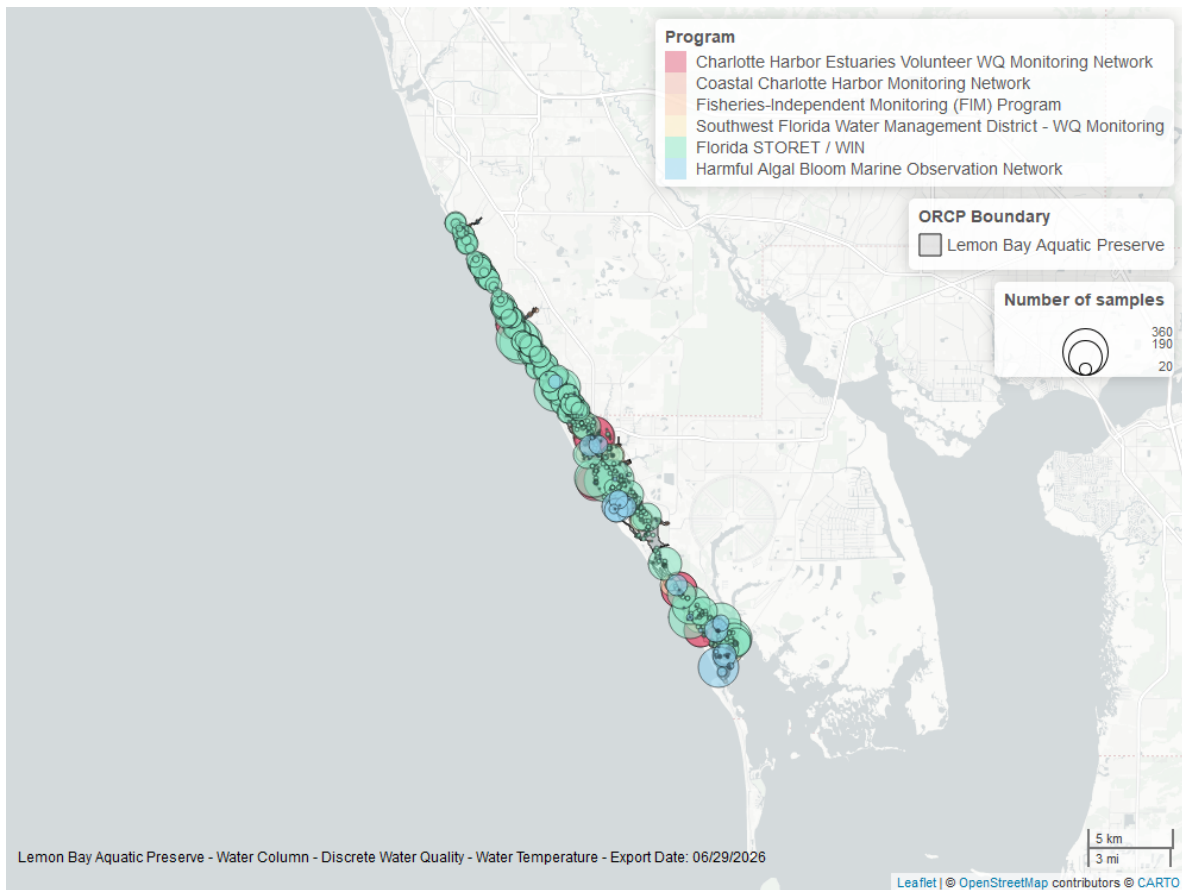


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

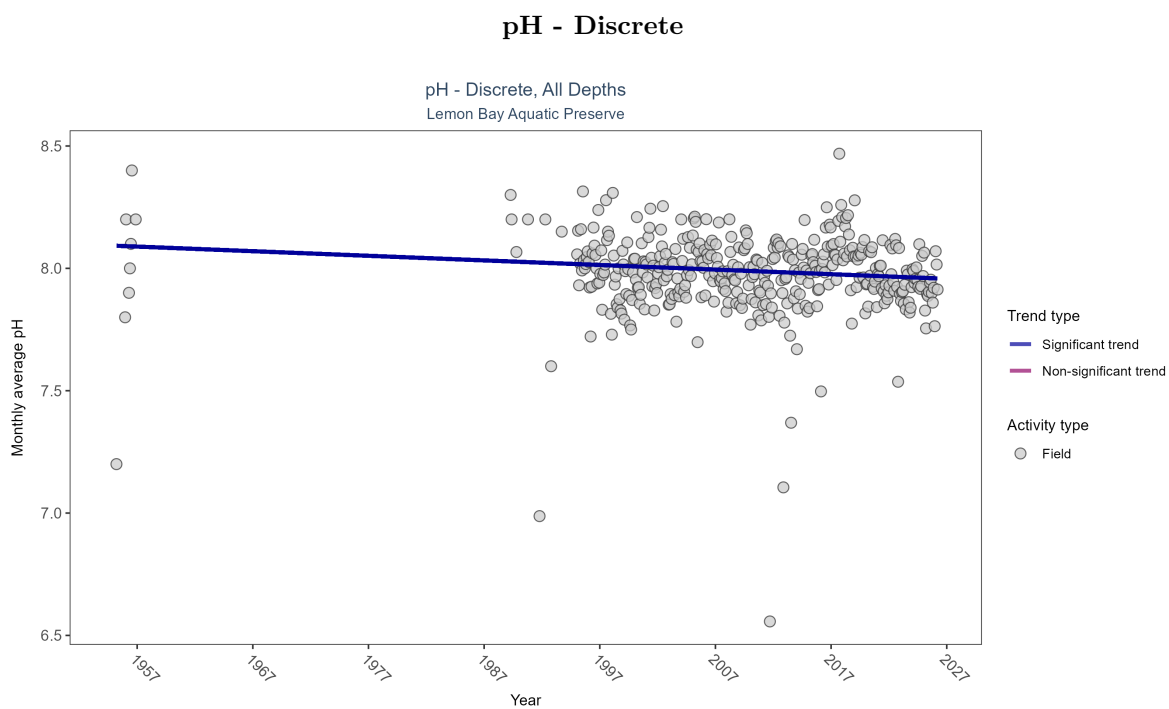


Figure 13: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 6: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	12321	39	1955 - 2026	8	-0.10076	8.09268	-0.00188	0.0041

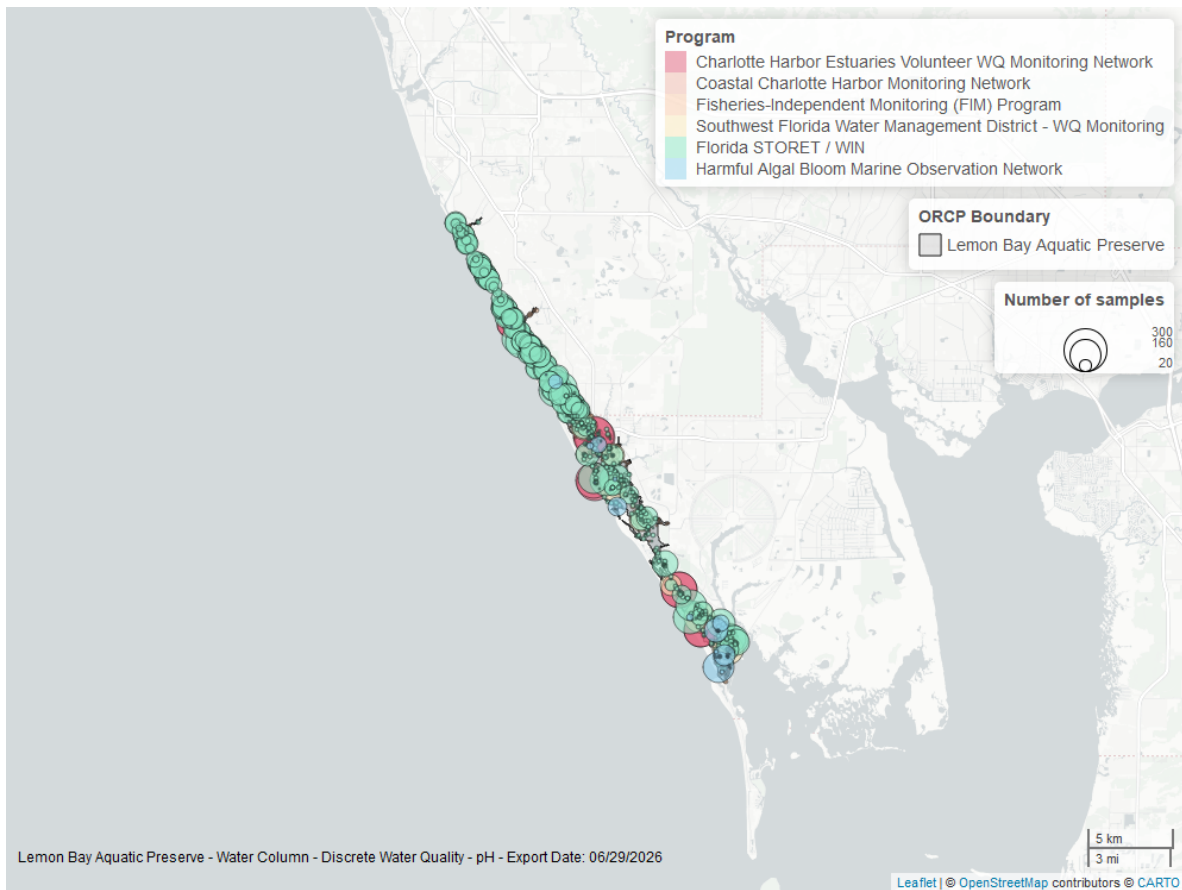


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

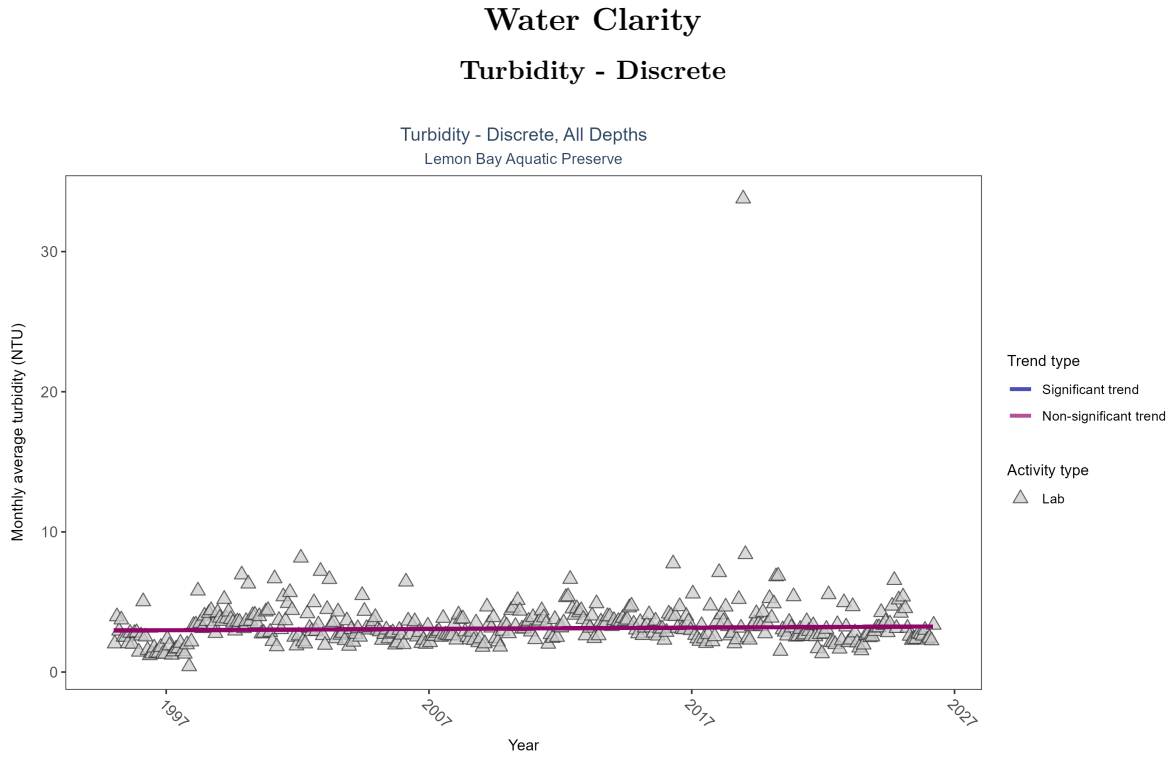


Figure 15: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 7: Turbidity showed no detectable trend between 1995 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	6245	32	1995 - 2026	2.6	0.04509	2.96421	0.00918	0.2146

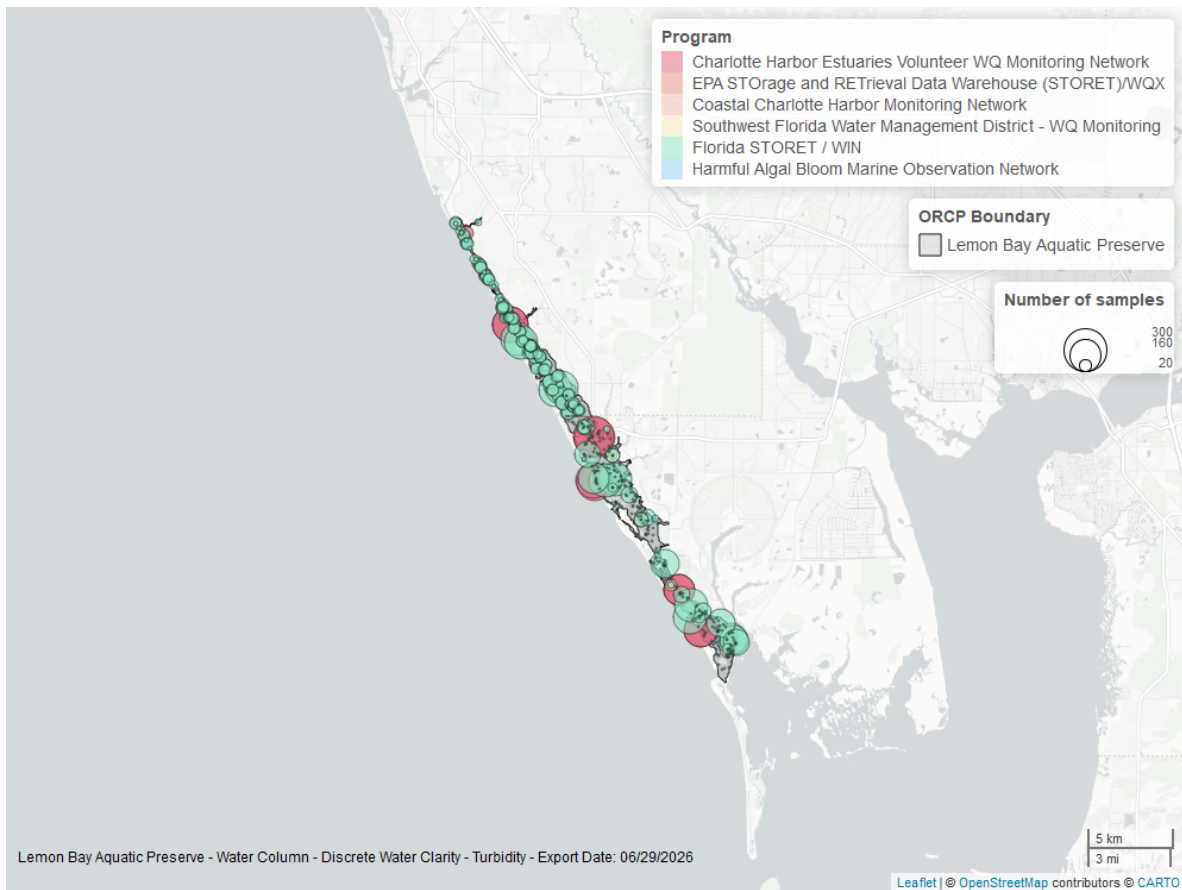


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

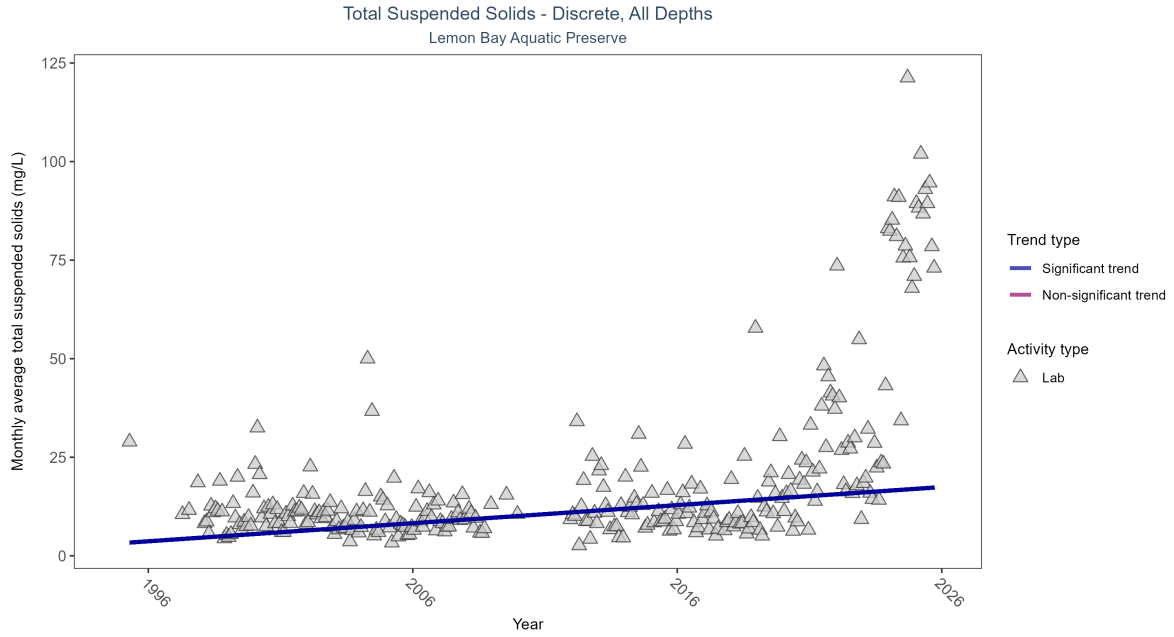


Figure 17: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 8: Monthly average total suspended solids increased by 0.46 mg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1276	29	1995 - 2025	10.8	0.3466	3.22453	0.45858	0

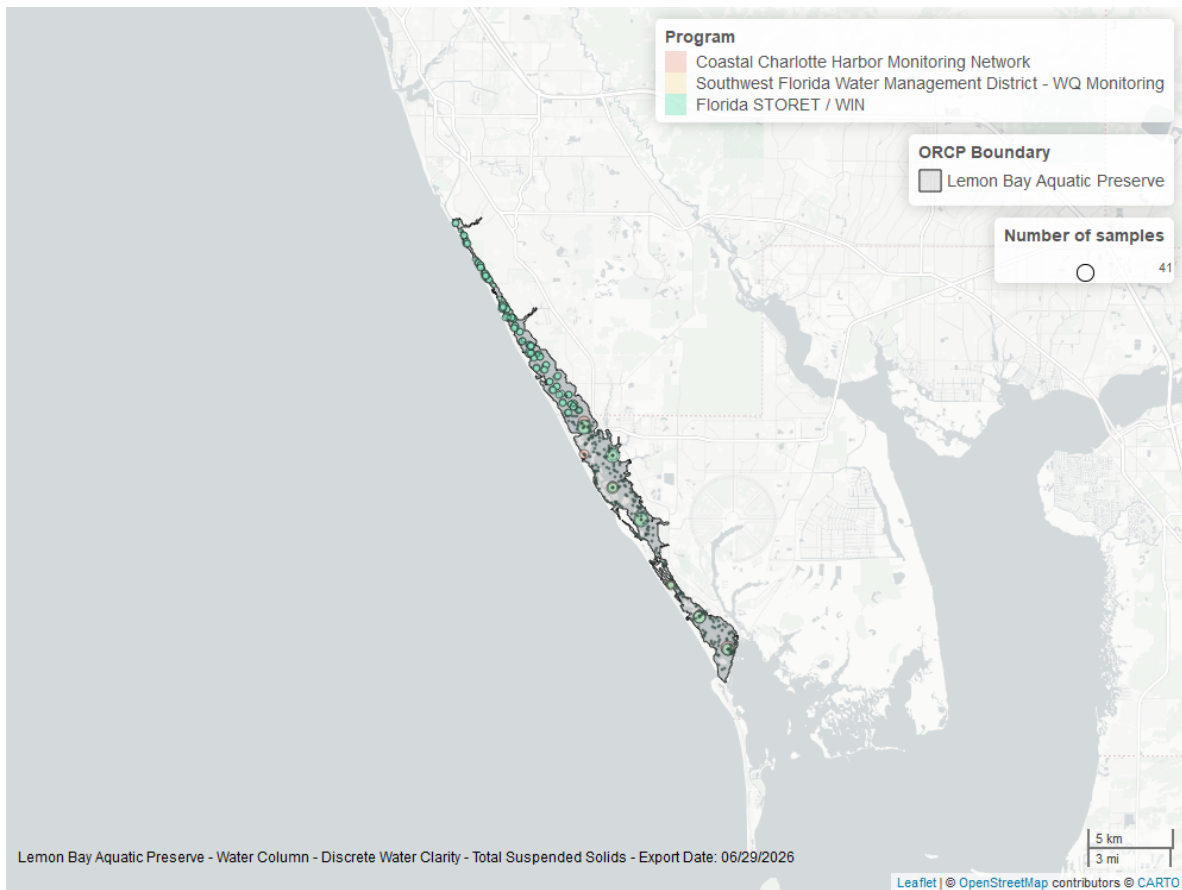


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

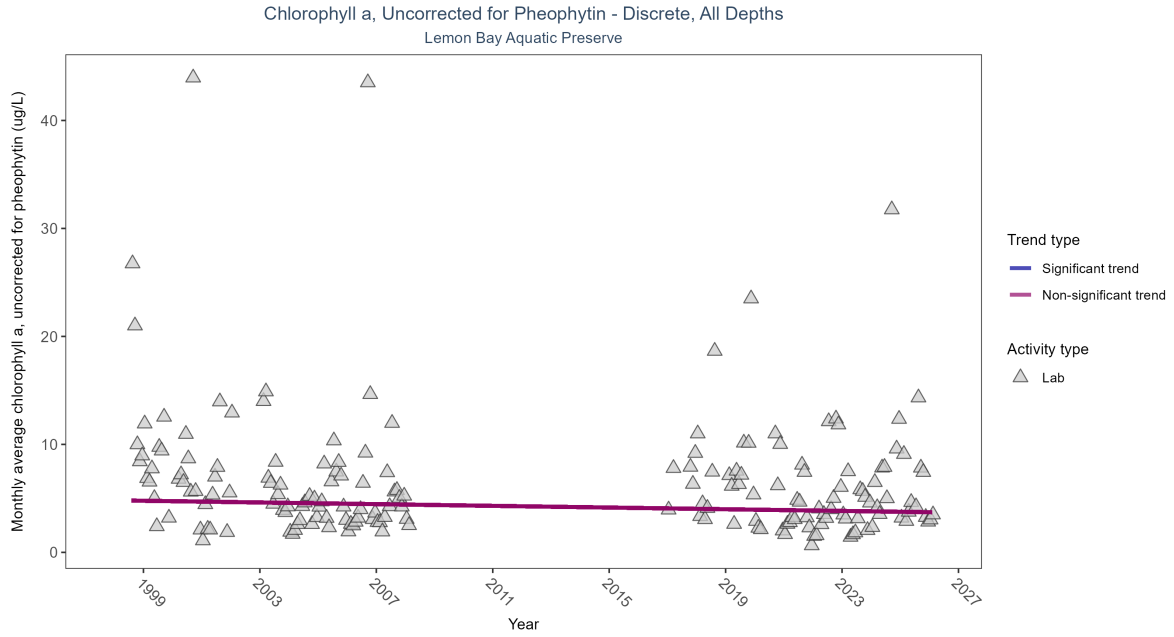


Figure 19: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 9: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 1998 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	802	21	1998 - 2026	4.07	-0.08698	4.81915	-0.03927	0.1325

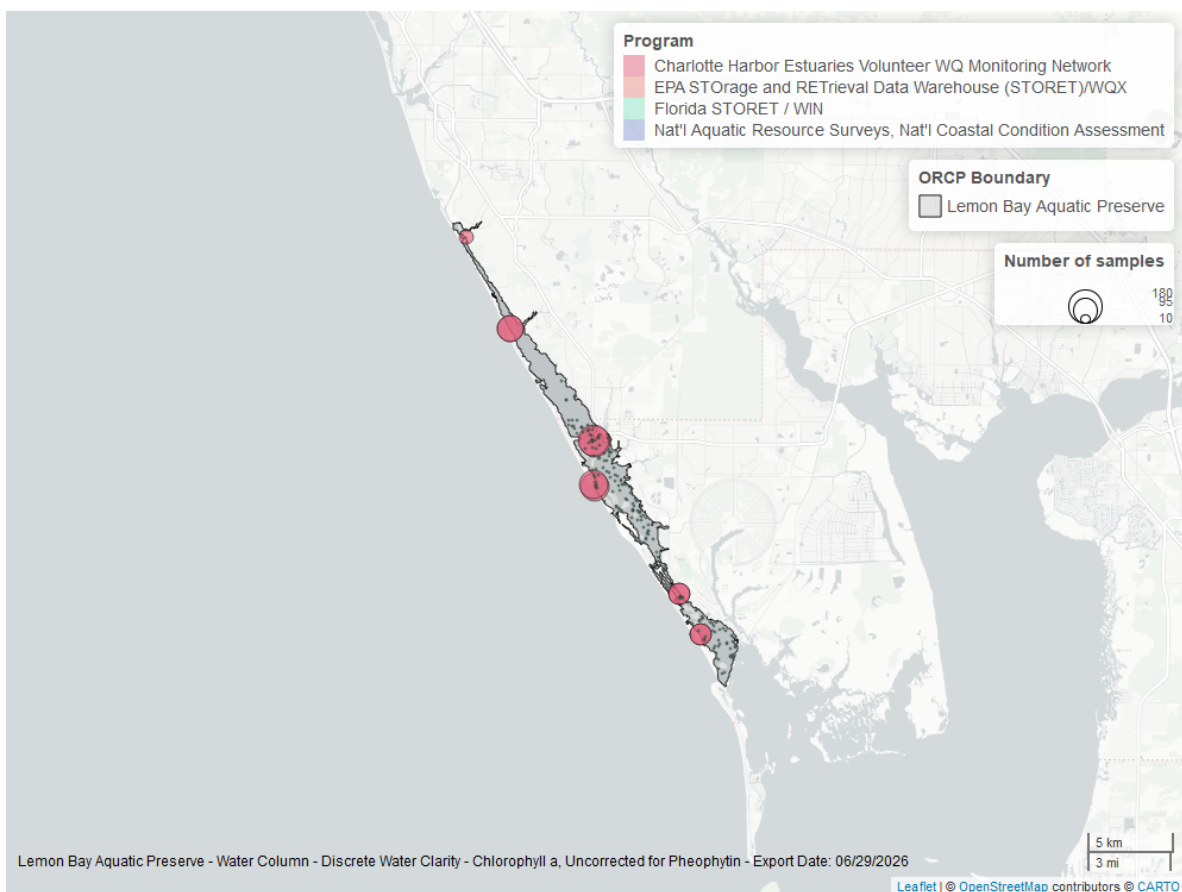


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

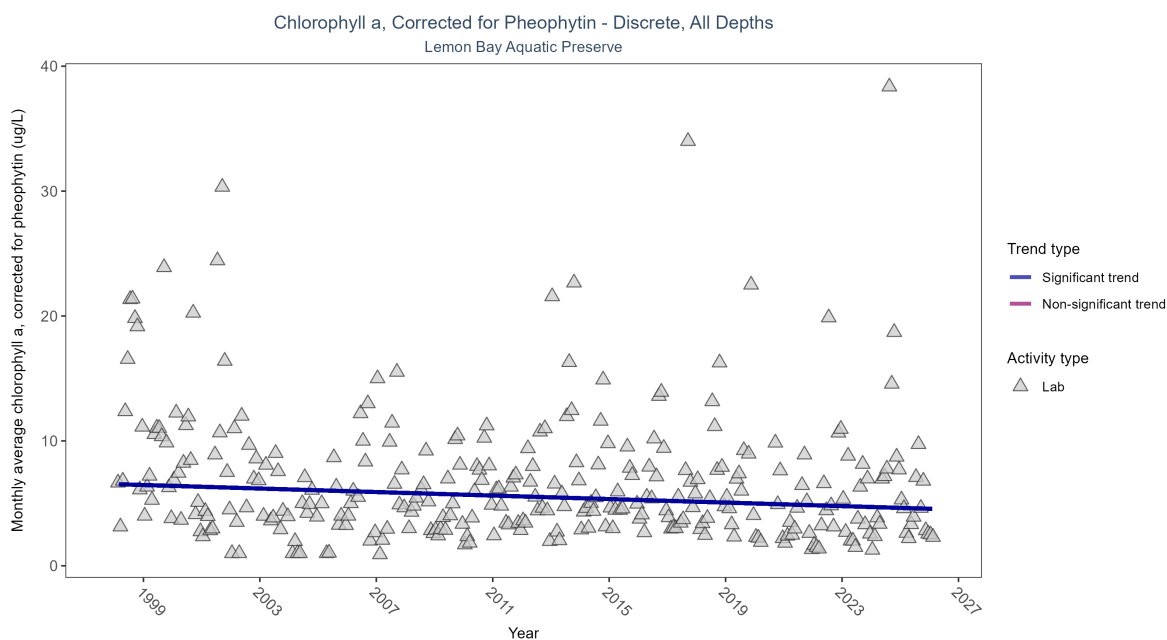


Figure 21: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 10: Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.07 ug/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	1308	29	1998 - 2026	4.84	-0.14519	6.53895	-0.07037	0.0003

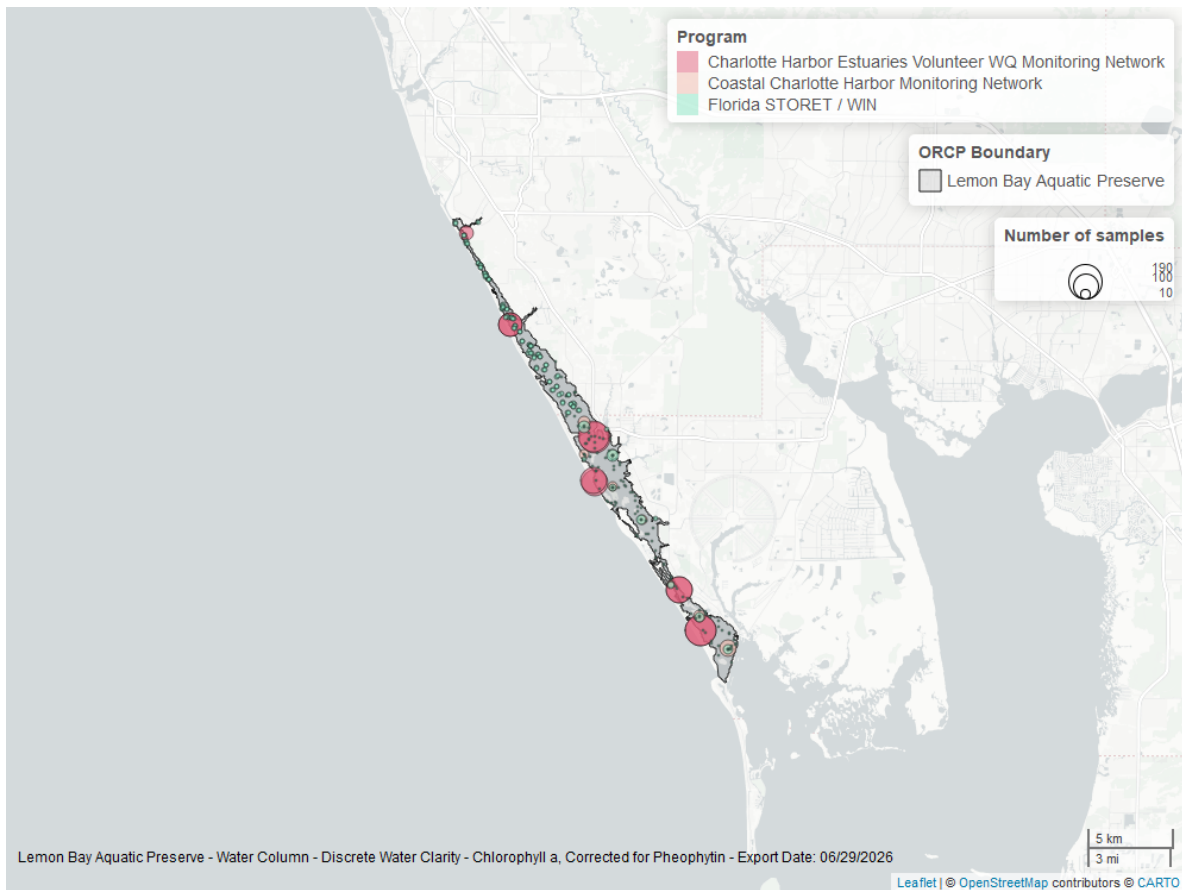


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

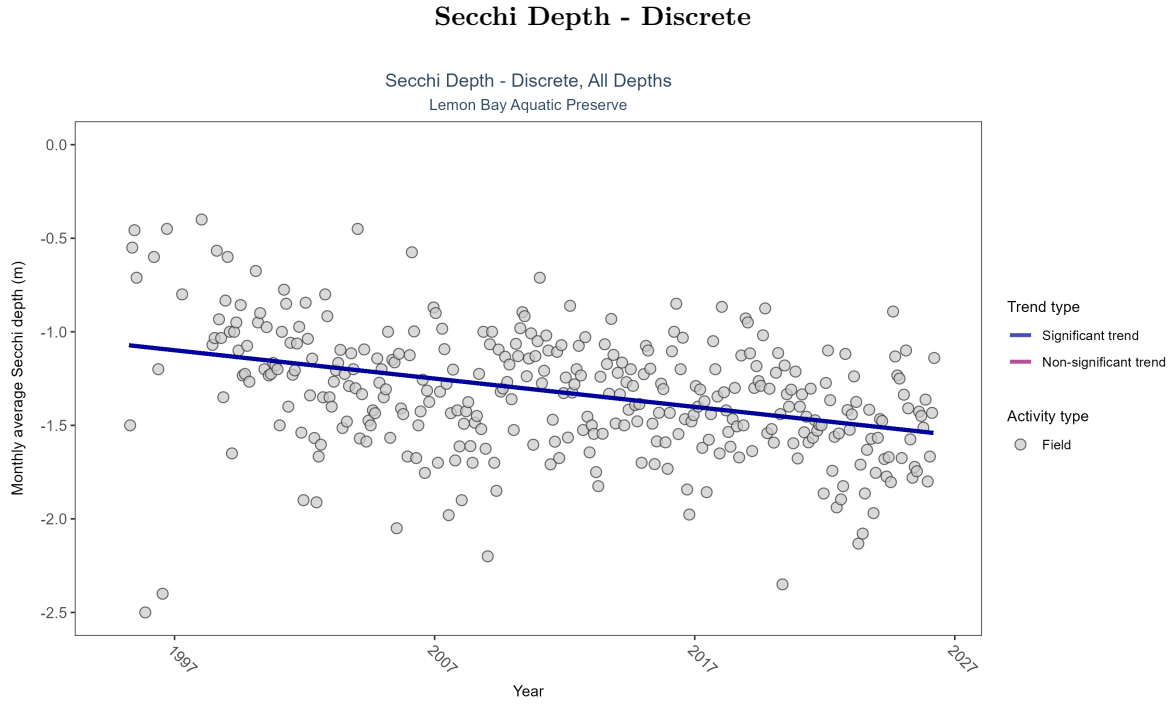


Figure 23: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 11: Monthly average Secchi depth became deeper by 0.02 m per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	3252	32	1995 - 2026	-1.3	-0.2901	-1.06829	-0.01515	0

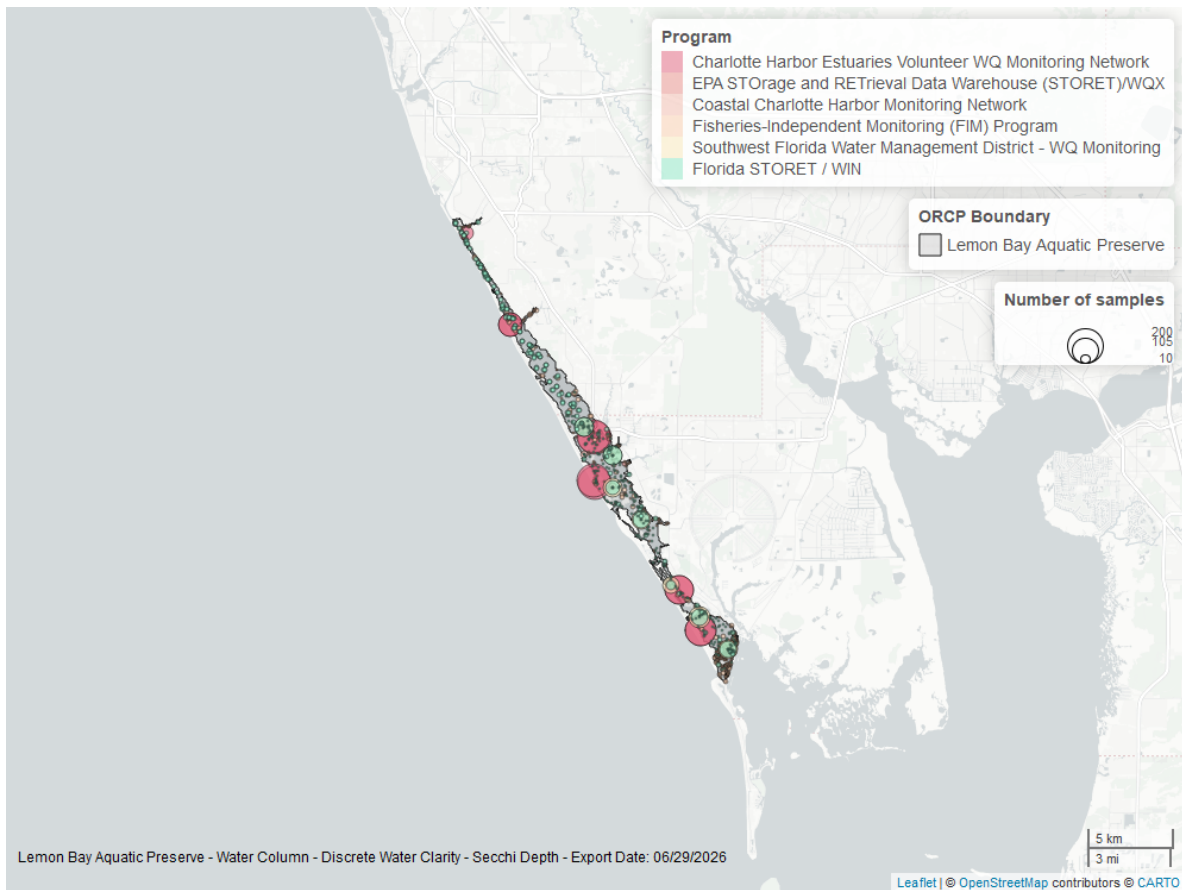


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

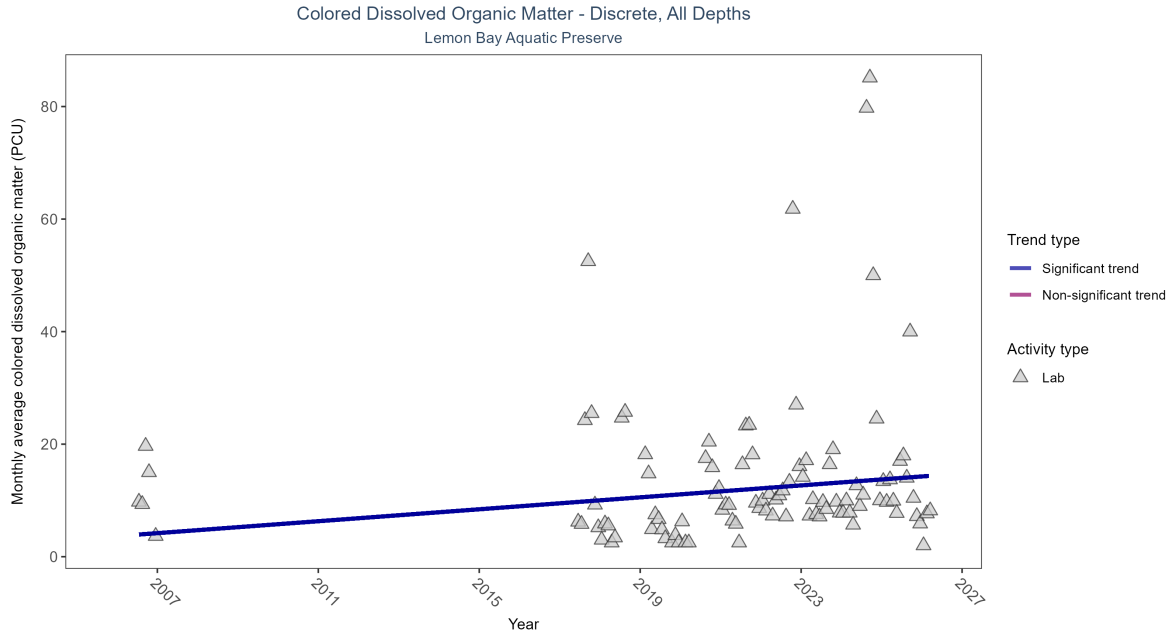


Figure 25: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 12: Monthly average colored dissolved organic matter increased by 0.53 PCU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	607	11	2006 - 2026	10	0.25794	3.65377	0.53028	0.004

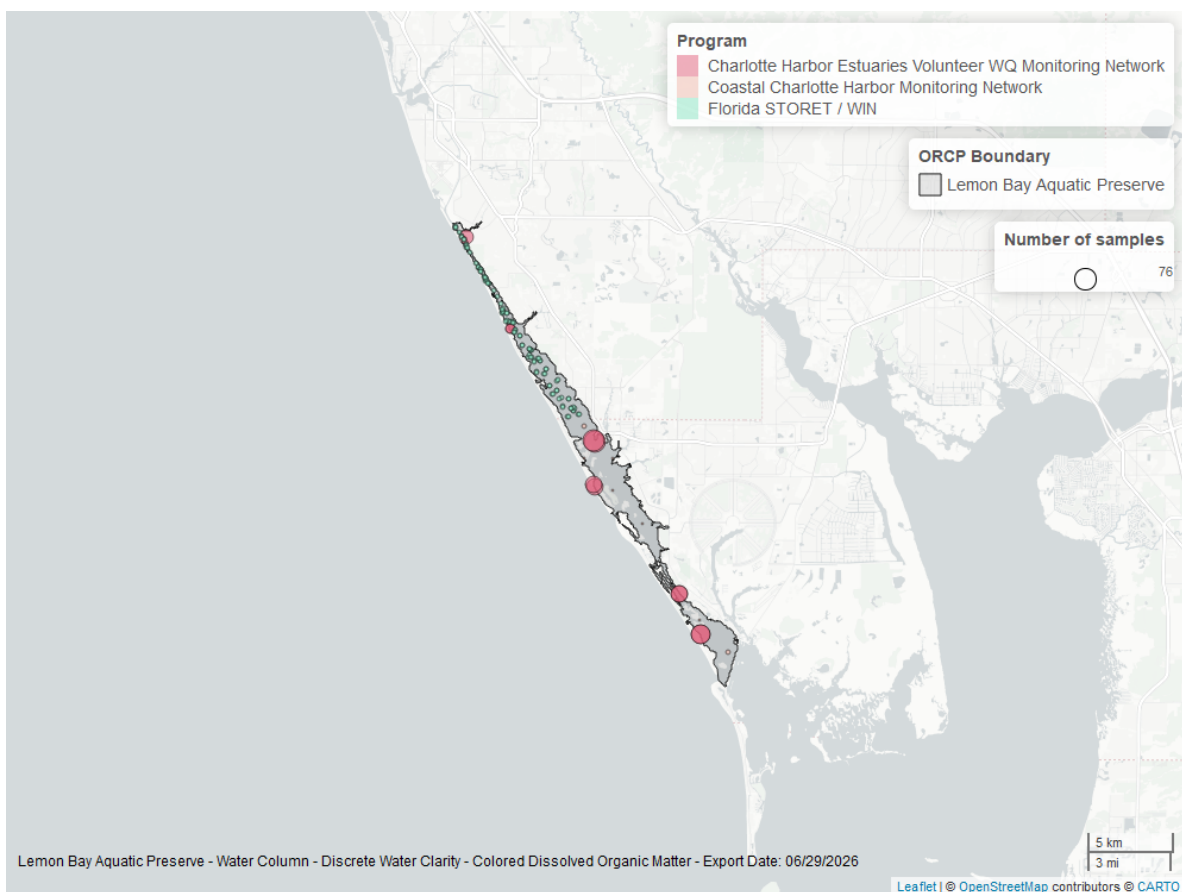


Figure 26: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.